



Team name: Telementors

Repository Link -

https://github.com/Telementors/Urban_flow_Analyzis.git

Demonstration Video –

https://youtu.be/jPG1d8y_U7E

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01.Executive Summary

Urban transportation systems generate large volumes of trip-level data containing information about travel times, locations, passenger counts, distances, fares, payment methods, and operational characteristics. However, raw transportation data may contain missing values, inconsistent records, abnormal observations, and complex spatial-temporal patterns that can limit the reliability of analytical and predictive systems. The Urban Flow Analytics Data Challenge focuses on transforming such raw operational data into useful insights and predictive solutions that can improve taxi operations.

This study develops a data-driven Urban Flow Analytics solution using the provided taxi trip dataset and geographical zone reference dataset. The taxi dataset provides trip-level information including pickup and drop-off timestamps, locations, passenger counts, distances, fare components, and other operational attributes, while the dataset provides geographical information that can be joined using location identifiers.

The proposed analytical pipeline focuses on four major areas: data quality assessment and exploratory data analysis, predictive modelling, spatial-temporal demand analysis, and business-oriented visualization. The preprocessing stage investigates data quality issues and prepares the raw records for downstream modelling. Feature engineering is used to transform temporal, categorical, numerical, and geographical information into useful analytical variables. The challenge specifically requires attention to negative fares, zero-distance trips with non-zero fares, zero passenger counts, invalid trip chronology, and unrealistic travel speeds.

For operational decision-making, the solution targets demand forecasting and spatial-temporal analysis. Demand forecasting can support fleet dispatching by predicting pickup volumes for future periods, while hotspot and origin-destination analysis can reveal important travel flows and areas of concentrated activity. The final analytical outputs are intended to transform raw transportation records into actionable information that can support operational efficiency, revenue improvement, customer experience, and data-driven management decisions.

Overall, the project demonstrates an end-to-end approach in which raw taxi data are systematically assessed, cleaned, transformed, analyzed, and used for predictive and spatial-temporal analytics. The resulting insights provide a foundation for more effective fleet positioning, demand planning, and management decision-making.

02.Introduction And Problem

Urban taxi networks produce millions of operational records describing individual journeys, including pickup and drop-off times, locations, travel distance, passenger counts, fares, and payment-related information. Although these records provide a valuable source for understanding urban mobility, the raw data is not necessarily ready for direct analytical use. Invalid observations, unusual values, inconsistent timestamps, and high-dimensional spatial-temporal patterns can affect the reliability of analytical results and predictive models. The Urban Flow Analytics challenge therefore requires the transformation of raw operational data into reliable information that can support improved taxi operations.

The first problem addressed in this study is **data quality**. The challenge identifies several important anomaly types, including negative fares, trips with zero distance but non-zero fares, zero passenger counts, drop-off timestamps occurring before pickup timestamps, and unrealistic travel speeds calculated from distance and trip duration. Each anomaly must be investigated and an appropriate treatment, such as filtering, removal, or imputation, should be justified and quantified.

The second problem is **predictive and operational analytics**. Taxi companies need to anticipate operational conditions rather than react only after demand occurs. Fleet dispatchers need information about where and when passengers are likely to request rides. The challenge therefore proposes forecasting pickup demand for major taxi zones over the next 24–72 hours to support better vehicle positioning and reduce unnecessary empty travel and passenger waiting time.

The third problem is **understanding spatial movement patterns**. Individual taxi records can be difficult to interpret when millions of trips are considered simultaneously. Spatial-temporal analysis and origin-destination clustering can aggregate these individual journeys into meaningful travel patterns, enabling the identification of high-demand areas and major movement flows across different periods of the day.

Therefore, the main objective of this work is to establish an end-to-end analytical pipeline that converts raw taxi trip data into reliable features, predictive outputs, and interpretable spatial-temporal insights. The resulting analysis is designed not merely to describe historical taxi activity but to support practical decisions related to fleet allocation, demand planning, operational efficiency, and customer service.

03. Dataset Description

The Urban Flow Analytics challenge provides two principal datasets in CSV format: the **Urban Flow Analytics Taxi Dataset** and the **Urban Flow Analytics Zone Dataset**. The taxi dataset contains trip-level operational records, whereas the zone dataset provides geographical reference information for taxi locations.

The taxi data are provided as monthly CSV files. The preprocessing notebook identifies monthly files covering **April 2025 to February 2026**, including April–December 2025 and January–February 2026. The files are processed individually before being combined for downstream analysis. Because the monthly files are large, the notebook initially reads only a small number of rows for structural inspection rather than loading complete files into memory. The notebook also demonstrates that the raw taxi records contain fields such as `provider_code`, `pickup_timestamp`, `dropoff_timestamp`, `rider_count`, `distance_miles`, `rate_class_id`, `offline_record_flag`, `origin_loc_id`, `dest_loc_id`, fare-related variables, and additional fee variables.

The **pickup and drop-off timestamps** provide the temporal dimension of the dataset and allow trip duration and time-based demand patterns to be analysed. `distance_miles` represents the recorded trip distance, while `rider_count` provides passenger information. Location identifiers, namely `origin_loc_id` and `dest_loc_id`, describe the origin and destination zones of each trip. Fare-related attributes include `base_fare`, `surcharge_misc`, `transit_tax`, `driver_tip_payment`, `toll_total`, `charge_total`, `zone_congestion_fee`, `Airport_fee`, and `congestion_relief_fee`.

The zone reference dataset contains `loc_id`, `borough_name`, `zone_name`, and `service_zone`. For example, the dataset maps location identifiers to geographical areas such as Queens, Bronx, Manhattan, and Staten Island, together with individual zone names and service-zone categories. This reference information enables location identifiers in the taxi records to be interpreted geographically and supports hotspot and origin-destination analysis.

Due to the size of the monthly taxi files, the preprocessing workflow uses chunk-based processing. Instead of loading the complete files into memory simultaneously, records are processed in manageable batches before being combined into the analytical dataset. This design improves memory efficiency while preserving the information required for subsequent exploratory analysis and modelling.

The resulting dataset provides three complementary dimensions for analysis: **temporal information** from journey timestamps, **operational and financial information** from trip and fare attributes, and **spatial information** from origin and destination zone identifiers. These dimensions form the basis for data-quality assessment, exploratory analysis, predictive modelling, demand forecasting, and spatial-temporal analytics.

05. Data Quality And Exploratory Data Analysis

A. Data Quality Assessment

Data quality assessment was performed before predictive modelling to determine whether the raw taxi records were suitable for analysis. The monthly files were first inspected separately to identify their structure, data types, representative values, and potential inconsistencies. The dataset contains both numerical and categorical variables together with timestamp fields, requiring different validation strategies. The notebook also examines the zone reference data separately before using it for geographical enrichment.

The challenge specifies five major anomaly categories that require explicit investigation: **negative fares, zero-distance trips with non-zero fares, zero passenger counts, drop-off timestamps earlier than pickup timestamps, and unrealistic travel speeds**. These checks are important because such records can distort descriptive statistics and introduce noise into predictive models.

A data-quality summary should be reported using the following format:

Data-quality issue	Detection rule	Treatment
Negative fare	<code>base_fare < 0</code>	Filter/remove
Zero distance with non-zero fare	<code>distance_miles = 0 and fare > 0</code>	Filter/remove
Zero passengers	<code>rider_count = 0</code>	[Treatment used]
Invalid chronology	<code>dropoff_timestamp < pickup_timestamp</code>	[Treatment used]
Unrealistic speed	Distance / trip duration above selected threshold	[Treatment used]

The above percentages should be calculated against the total number of raw records rather than against only the affected subset. This provides a transparent measure of the magnitude of each quality problem, as required by the challenge.

Missing-value analysis was also considered as part of the initial data inspection. The preprocessing workflow calculates missing-value statistics for the available fields before the datasets are combined. This allows columns with incomplete observations to be identified before feature transformation and modelling.

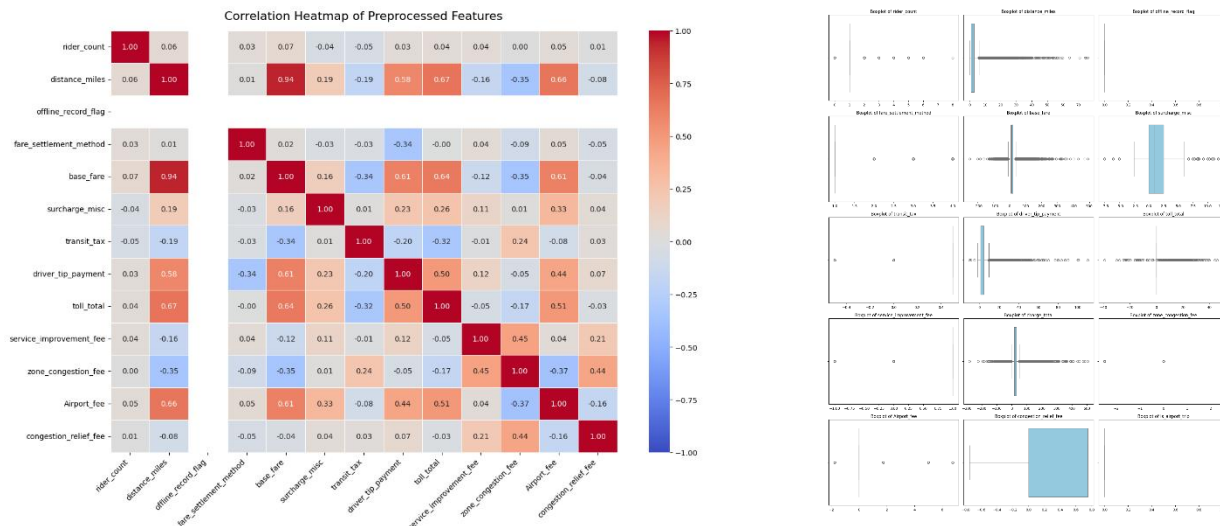
B. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the statistical characteristics and relationships within the taxi data. Representative records from the monthly files show variation in trip distance, passenger count, base fare, tips, total charges, and geographical origin and destination. For example, the inspected records demonstrate distances ranging from short urban trips to longer journeys and variation in fare components across individual trips.

The temporal variables provide an important basis for analyzing taxi demand. Pickup and drop-off timestamps can be transformed into features such as hour, day, weekday, and month to identify recurring demand patterns. These temporal patterns are particularly relevant because the challenge requires pickup-volume forecasting for future periods and highlights the importance of understanding demand changes across time.

The spatial variables were examined using origin and destination location identifiers. The zone dataset enables these identifiers to be mapped to boroughs, zone names, and service-zone categories. This provides the foundation for identifying high-activity locations and analyzing origin-destination movement patterns.

Numerical distributions and relationships were also investigated using descriptive statistics, boxplots, and correlation analysis. The purpose of these analyses was to identify skewed variables, potential extreme observations, highly related variables, and possible redundancy before modelling. The preprocessing workflow uses representative samples for computationally efficient numerical analysis while retaining the full dataset for the subsequent processing pipeline.



The EDA provides several analytical foundations for subsequent stages. Temporal variation supports demand forecasting, geographical variation supports hotspot and origin-destination analysis, and numerical relationships help identify useful predictive variables and redundant features. These findings are therefore used to guide the subsequent preprocessing, feature engineering, predictive modelling, and dashboard development stages.

06. Preprocessing And Feature Engineering

The raw Urban Flow Analytics Taxi Dataset consists of multiple monthly CSV files containing trip-level information such as pickup and drop-off timestamps, passenger count, travel distance, location identifiers, fare components, payment information, and operational attributes. Since the monthly files are large, a memory-efficient preprocessing pipeline was developed using chunk-based processing. The monthly taxi files were first inspected individually, and missing-value statistics were calculated for each column before combining the datasets. The files were subsequently combined using 100,000-row chunks to avoid excessive memory consumption.

To preserve the temporal nature of taxi demand and trip behavior, the data were divided using a rolling 70:30 strategy rather than a random split. Groups of monthly files were processed sequentially, with three months allocated to training and the following month allocated to testing. This approach reduces temporal data leakage and provides a more realistic evaluation of the predictive models.

Feature engineering was performed to improve the representation of operational and geographical characteristics. An **is_airport_trip** binary feature was generated from `Airport_fee`, where trips with an airport fee greater than zero were assigned a value of 1 and other trips were assigned 0. The zone reference dataset was also transformed by replacing missing categorical values with an Unknown category. Low-cardinality variables such as `borough_name` and `service_zone` were converted using one-hot encoding, while `zone_name` was represented using frequency encoding to reduce the dimensionality associated with categorical location information.

Categorical preprocessing was additionally applied to `offline_record_flag`, mapping N and n to 0 and Y and y to 1, with missing values treated as 0. Numerical relationships were examined using Pearson correlation matrices and boxplots on representative 200,000-row samples to identify highly correlated and potentially redundant variables.

For numerical feature normalization, `StandardScaler` was fitted **only on the training data** using incremental `partial_fit`, preventing information from the test set from influencing the transformation. The selected numerical variables included `distance_miles`, `base_fare`, `driver_tip_payment`, `toll_total`, and `surcharge_misc`. Records with non-positive `base_fare` or `distance_miles` were filtered during preprocessing. Highly redundant variables were subsequently removed; `charge_total` and the engineered `is_airport_trip` feature were initially excluded during transformation, while further correlation analysis identified `base_fare` as highly redundant with `distance_miles`. Constant features were also considered for removal. The resulting processed datasets were stored as separate training and testing files for downstream modelling.

Overall, the preprocessing pipeline converts the heterogeneous raw taxi records into a consistent, scaled, encoded, and leakage-aware dataset suitable for predictive modelling, demand forecasting, and spatial-temporal analysis.

07. Fare Amount Prediction

The primary objective of the fare amount prediction task is to accurately estimate the foundational cost of a taxi trip based on various operational, temporal, and spatial features. The specific target variable for this predictive modeling exercise is the base fare, which represents the core cost of the journey before additional surcharges, taxes, tolls, or tips are applied. Predicting this metric accurately is crucial for ensuring pricing transparency, optimizing revenue management, and understanding the fundamental financial dynamics of urban taxi operations. By isolating the base fare, the model focuses on the intrinsic value of the trip's distance, duration, and geographic routing.

Before modeling, an extensive exploratory data analysis was conducted to understand the underlying distributions and relationships within the trip data. The analysis revealed strong positive correlations between trip distance and base fare, alongside noticeable temporal variations where fares fluctuated based on the hour of the day and day of the week, reflecting peak commuting and weekend travel behaviors. Spatial analysis using origin and destination location identifiers highlighted that specific zones inherently commanded different base rates due to varying distances or zone-specific pricing rules. Furthermore, examining trip duration alongside distance helped identify anomalous records, such as unusually long trips covering short distances, which were subsequently flagged for cleaning to prevent model distortion.

Developing a robust fare prediction model presented several significant technical and analytical challenges. The most immediate hurdle was the sheer volume of the dataset, comprising over 12.2 million trip records, which frequently caused memory exhaustion in the cloud computing environment. Additionally, the raw data contained numerous inconsistencies, including negative fares and zero-distance trips with non-zero charges, which threatened the integrity of the predictive models. From a modeling perspective, initial experiments using a standard Linear Regression algorithm yielded highly unsatisfactory results, producing an R-squared value near zero and a high Root Mean Squared Error. Residual analysis confirmed that the relationship between the engineered features and the base fare was inherently non-linear, rendering simple linear approaches entirely ineffective for this complex urban dataset.

To overcome the memory constraints, a highly optimized, chunk-based data processing pipeline was engineered. The dataset was read and processed in manageable batches of 500,000 rows, applying rigorous data cleaning to filter out invalid fares and distances, while simultaneously downcasting numeric data types to minimize the RAM footprint. To address the non-linear complexities of fare determination, the modeling strategy was pivoted from linear regression to advanced ensemble methods, specifically the Random Forest Regressor. This algorithmic shift allowed the model to capture intricate, non-linear interactions between spatial zones, temporal patterns, and trip distances without requiring exhaustive manual feature engineering.

08. Trip Duration Prediction

Target Definition

The primary objective of this project is to accurately predict `travel_duration` (measured in total elapsed seconds), derived directly as the time difference between `dropoff_timestamp` and `pickup_timestamp`. By forecasting trip duration rather than relying solely on spatial metrics, the system enables precise time estimates for riders, improves driver scheduling efficiency, and prevents target leakage caused by downstream financial indicators such as total fare charges.

Exploratory Data Analysis (EDA) & Workflow

The dataset consists of massive ride-hailing transactional logs containing spatial identifiers (`origin_loc_id`, `dest_loc_id`), travel distances (`distance_miles`), temporal stamps, rider counts, and fare breakdowns. Initial exploratory steps revealed strong temporal non-stationarity and cyclical patterns in demand; specifically, trip durations vary significantly depending on peak traffic hours and day-of-week trends. Feature engineering focused on extracting temporal representations—`pickup_hour_of_day`, `dropoff_hour_of_day`, `pickup_day_of_week`, and `dropoff_day_of_week`—while eliminating non-predictive administrative features (such as `provider_code`) and post-trip financial factors (such as `base_fare`, `surcharge_misc`, `toll_total`, and `charge_total`) to ensure realistic real-time inference.

Challenges Faced

- **Memory Constraints & Scale:** Processing multi-gigabyte transactional files exceeded available system RAM when attempting standard full-dataset loading into Pandas DataFrames.
- **Mixed Data Types & Unlabeled Headers:** Raw data files initially lacked explicit column headers and contained mixed structural dtypes across categorical flags.
- **Data Leakage Risk:** Financial metrics (e.g., base fare, tip amount, total charge) are determined *after* or during a trip, creating an artificial data leakage path if included as predictors.

Architectural Solutions & Modeling Results

To overcome memory limitations, a chunk-based streaming pipeline (`chunksize=200,000`) was engineered using Pandas and Python's garbage collection (`gc.collect()`). Categorical variables like `offline_record_flag` were dynamically encoded into compact 8-bit integer formats (`int8`), and numerical features were downcast to 32-bit floats (`float32`), reducing the dataset's memory footprint dramatically. To mitigate data leakage, all fare-related and post-trip features were systematically removed during chunk extraction.

For predictive modeling, an **XGBoost Regressor** was deployed utilizing histogram-based tree optimization (`tree_method='hist'`) for fast GPU execution and memory efficiency. Trained with early stopping over 500 boosting rounds, the final model achieved strong predictive capability on the validation set:

- **Validation RMSE:** 647.38 seconds (~10.7 minutes)
- **Validation MAE:** 203.82 seconds (~3.4 minutes)
- **Validation R² Score:** 0.8248 (~82.5% variance explained)

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09.Demand Forecasting

A. Operational Problem and Business Objective

Urban taxi operations are strongly affected by spatial and temporal imbalances between taxi supply and passenger demand. When drivers are unable to anticipate where demand will occur, idle vehicles may cruise without passengers, increasing fuel consumption and reducing revenue, while high-demand areas may experience longer passenger waiting times. To address this challenge, **UrbanFlow Demand Intelligence** was developed as a predictive dispatching engine capable of forecasting hourly pickup demand across **261 taxi zones** for a future horizon of up to **72 hours**. This supports proactive fleet repositioning by allowing dispatchers to identify expected demand concentrations before they occur. The approach directly addresses the Fleet Dispatcher requirement of predicting pickup volumes over the next 24–72 hours.

B. Data Formulation and Spatial-Temporal Aggregation

Approximately **11.8 million validated taxi trip records** were transformed into a structured zone-hour demand dataset. Each pickup timestamp was floored to an hourly interval using `pickup_timestamp.dt.floor('h')` and aggregated by `origin_loc_id`. The resulting target variable, **pickup_count**, represents the number of pickups occurring within a specific taxi zone during a given hour. This aggregation produced **353,963 zone-hour observations** across 261 zones, reducing trip-level noise while retaining important spatial and temporal demand patterns.

The forecasting feature set consists primarily of deterministic temporal and spatial variables. `hour_of_day` (0–23) captures daily demand cycles, while `day_of_week` (0–6) represents weekly seasonality. An `is_weekend` indicator further distinguishes weekday commuting behaviour from weekend travel patterns. The `origin_loc_id` feature represents the spatial component of demand. Rather than relying on recursive lag variables, the forecasting formulation uses these deterministic time and location attributes to generate future zone-hour predictions independently. This design

avoids recursive error propagation and enables direct multi-step prediction across the complete 72-hour horizon.

C. Model Architecture and Optimization

An **XGBoost regression model** was selected to capture nonlinear relationships between temporal patterns, taxi zones, and pickup demand. The model uses the `reg:squarederror` objective to minimize squared prediction errors and the histogram-based tree construction method (`tree_method='hist'`) to improve computational efficiency.

The dataset was divided into **80% training (283,170 zone-hours)** and **20% unseen testing (70,793 zone-hours)** using `random_state=42`. The principal model configuration consisted of **600 estimators**, a maximum tree depth of **8**, a learning rate of **0.10**, and both row and feature subsampling ratios of **0.80**. Early stopping with a patience of 50 rounds was incorporated to reduce unnecessary boosting iterations and control overfitting.

D. Empirical Evaluation

The final XGBoost model achieved strong predictive performance on the unseen test set, obtaining an **RMSE of 14.90**, **MAE of 6.46**, and **R² of 0.9575**. The coefficient of determination indicates that approximately **95.75% of the observed variation in hourly zone-level demand** is explained by the model.

The **MAE of 6.46 pickups/hour** provides an intuitive operational interpretation: predictions differ from observed demand by approximately six to seven pickups per zone-hour on average. The **RMSE of 14.90** gives greater weight to larger prediction errors, making it useful for identifying substantial under- or over-predictions during demand surges. The combination of MAE, RMSE, and R² therefore provides complementary information about average prediction accuracy, sensitivity to larger errors, and overall explanatory performance. These metrics are also consistent with the challenge's recommendation to evaluate regression models using RMSE, MAE, and R².

E. Production Inference and 72-Hour Deployment

For operational forecasting, the trained booster processed **372,755 inference records in approximately 4.66 seconds**, corresponding to more than **80,000 predictions per second**. A future timestamp grid was dynamically generated for all 261 active zones, producing **18,792 forecast points** for the 72-hour horizon. Forecast outputs are organized into three operational tiers: **0–24 hours** for tactical fleet dispatching, **25–48 hours** for shift and resource planning, and **49–72 hours** for strategic fleet balancing.

The forecasting pipeline is containerized using **Python 3.11 and Docker**, enabling reproducible execution and automated generation of forecast outputs. The resulting predictions are connected to a **Tableau dashboard**, where actual and predicted demand can be compared through time-series visualizations, peak-demand warnings, and dynamic Top-10 hotspot rankings. Consequently, the forecasting component converts historical trip data into forward-looking operational intelligence,

supporting proactive vehicle allocation, improved fleet utilization, and reduced passenger waiting time. This aligns with the challenge requirement that the final solution transform analytical results into practical business decisions.

10.Hotspot & OD Flow Analysis

Target & Objectives

The primary target of this implementation is to forecast high-resolution spatial and temporal urban movement through two simultaneous tasks: predicting hotspot traffic (hourly pickup and drop-off volumes per zone) and predicting Origin-Destination (OD) flow matrices (hourly trip volume completions between zone pairs). The system models localized temporal dependencies—capturing short-term closeness, daily periodic patterns, and weekly trends—alongside temporal calendar indicators. The overarching objective is to deliver precise, end-to-end spatiotemporal forecasting for operational urban planning while strictly maintaining a low memory footprint suitable for restricted hardware environments.

Core Challenges

The main engineering bottleneck in scaling spatiotemporal residual networks is the severe memory consumption associated with dense spatial matrix calculations. Processing high-dimensional zone-to-zone OD flow tensors ($1 \times Z \times Z$) alongside dual-channel hotspot vectors ($2 \times 1 \times Z$) across long historical time-series easily exceeds standard GPU VRAM and system RAM limits. Additional operational challenges include handling massive raw trip logs without loading full datasets into memory, avoiding data leakage during scaling, managing incomplete timestamp edge cases across multi-year trip archives, and ensuring model recovery across long-running training routines without losing accumulated epoch states.

Resolution Strategy

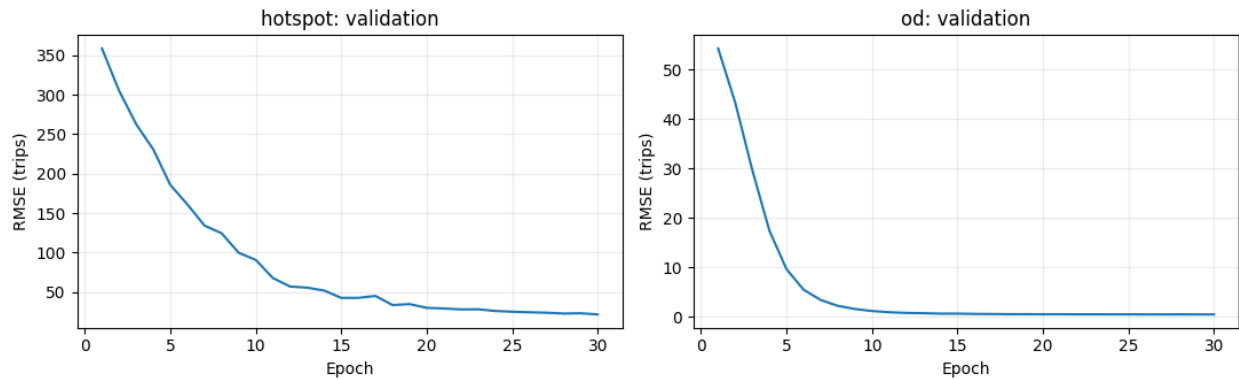
To overcome memory constraints, the pipeline uses disk-bounded processing and an optimized model architecture. System RAM is preserved by processing raw trip data in streaming chunks of 25,000 rows, selecting only essential columns, and building sparse count tables on local disk using an embedded SQLite engine with a 32 MiB bounded cache. GPU VRAM overhead is minimized by storing dense hourly records as raw float32 binary files and reading only requested time steps via custom disk-seeking file handlers (HourStore), avoiding pre-expanded lag arrays in memory.

Model capacity is trimmed to 16 hidden channels and 2 residual blocks per branch, paired with strict batch sizes (batch size of 1 for OD flow and 4 for hotspots) combined with CUDA

Automatic Mixed Precision (AMP). Data integrity and reproducibility are enforced by deriving zone vocabularies and min/max normalization parameters solely from training split bounds before applying a tanh activation layer, while atomic checkpointing allows robust pause and resume functionality.

Exploratory Data Analysis & System Bounds

The system scanned over 11.79 million raw records to map observed time horizons and extract active zone identifiers. Exploratory checks established a continuous evaluation window covering April 2025 (2025-04-01 through 2025-05-01), allocating 168 warm-up hours for temporal lag structures before partitioning the target timeline into chronological splits: 386 training target hours, 83 validation target hours, and 83 testing target hours. Filtering on valid geographic regions identified 260 active pickup and drop-off zones across the training horizon.



During the primary aggregation pass, 3,705,073 drop-off events and 3,706,120 pickup events were retained for hotspot modelling, while 3,671,155 completed, non-voided trips with positive durations were mapped to the target OD flow matrices. The total dense hourly count structures occupied only 0.186 GiB of local disk storage. Peak system memory utilization remained stable at approximately 4.11 GiB of RAM on a CUDA-enabled GPU runtime, proving the pipeline's memory efficiency.

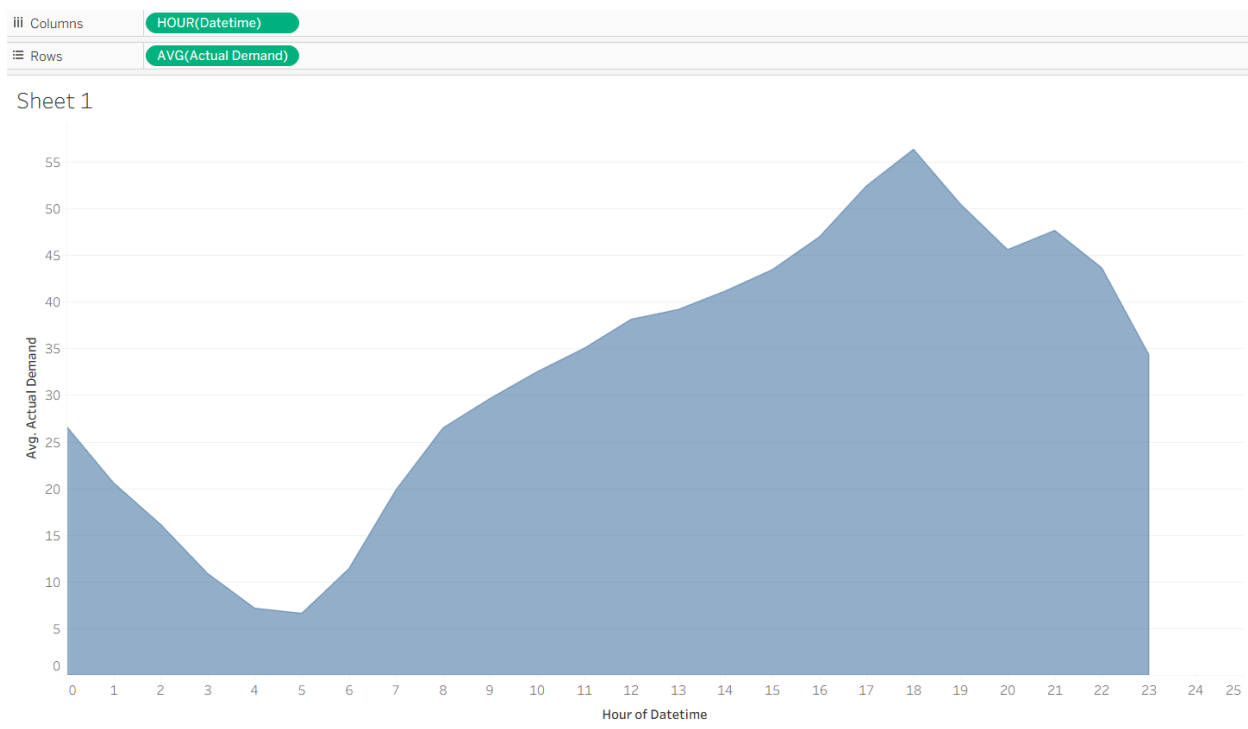
Bonus Track:

12.Business Dashboard

A business-oriented dashboard was developed using **Tableau** to present the outputs of the Demand Forecasting component in an interactive and decision-support format. The trained XGBoost model generated the future demand predictions, which were exported as a forecast dataset and subsequently connected to Tableau for visualization. The dashboard focuses specifically on **hourly pickup-demand forecasting across taxi zones**, allowing users to examine expected demand patterns over the upcoming 72-hour horizon.

The dashboard provides **actual versus predicted demand trends**, enabling comparison between historical pickup volumes and model forecasts. It also highlights **peak-demand periods** and identifies the **Top-10 high-demand taxi zones**, helping fleet managers determine where additional vehicles may be required. Interactive filtering allows users to explore demand according to zone and forecast period, making the model outputs easier to interpret for operational decision-making.

This dashboard transforms the machine-learning predictions into an easily interpretable business interface, supporting proactive fleet positioning and operational planning. The use of an interactive dashboard also supports the Datathon requirement that analytical results should help management understand the identified problem and make informed business decisions.



13. Business Recommendations

1. Proactive Fleet Dispatch and Idle-Time Reduction

The **72-hour Demand Forecasting model** provides fleet managers with advance visibility of expected pickup demand across taxi zones. Instead of relying entirely on real-time demand, vehicles can be proactively positioned in zones where higher demand is expected. This can help improve fleet utilization, reduce unnecessary cruising, and reduce passenger waiting times. The forecast can therefore support both short-term vehicle repositioning and operational shift planning.

2. Demand-Based Revenue Optimization

Forecasted demand patterns can also support revenue-oriented operational decisions. Identifying periods and zones with consistently high expected demand allows the company to allocate more vehicles to those locations and maximize the number of completed trips. Demand forecasts can additionally support future pricing strategies by providing an indication of expected demand intensity before peak periods occur. However, the current system should be considered a **decision-support mechanism rather than a direct dynamic-pricing engine**.

3. Improved Customer Service and Availability

The demand forecasting system can improve customer experience by helping ensure that sufficient vehicles are available in high-demand locations. Forecast information can be incorporated into a ride-hailing or fleet-management application to identify upcoming demand peaks and allocate vehicles accordingly. This proactive allocation has the potential to reduce vehicle shortages in busy zones and improve service availability during peak periods.

4. Urban Mobility and Origin-Destination Planning

The combination of demand forecasting and spatial analysis provides useful information for understanding urban mobility patterns. Frequently occurring high-demand zones and major origin-destination flows can be used to support decisions related to designated pickup areas, fleet staging locations, and transport planning. These insights can also assist transportation authorities and city planners in identifying areas where additional mobility infrastructure or improved traffic management may be beneficial.

6. Interactive Management Decision Support

The **Tableau Demand Forecasting Dashboard** converts the machine-learning predictions into an interactive business interface. Actual and predicted demand can be compared, peak-demand periods can be identified, and high-demand zones can be ranked using the forecast outputs. This allows operations managers and business decision-makers to move from **historical analysis to forward-looking planning**, supporting faster and more informed fleet-management decisions.

14.Limitations

1. Data and Environmental Limitations

Lack of Exogenous Features:

The current demand forecasting model primarily uses historical taxi demand together with temporal and spatial features. External factors such as real-time weather conditions, public events, traffic incidents, and transportation disruptions are not included. Consequently, sudden demand changes caused by unexpected external events may not be fully captured by the forecasting model.

Limited Historical Time Window:

The available dataset covers a one-year period from **April 2025 to March 2026**. Therefore, longer-term changes in travel behaviors and urban transportation patterns may not be fully represented in the training data.

2. Spatial and Temporal Granularity Limitations

Zone-Level Spatial Representation:

The forecasting system operates at the taxi-zone level using originalistic rather than exact GPS coordinates. As a result, demand variations within individual zones cannot be represented at street or block level.

Hourly Demand Aggregation:

Pickup demand is aggregated into one-hour intervals. This provides a practical balance between forecasting detail and computational efficiency; however, very short-term demand spikes occurring within an hour may not be individually captured.

3. Forecasting Model Limitation

Deterministic Multi-Horizon Forecasting:

The 24–72-hour forecasting process uses future-known temporal and spatial features such as `hour_of_day`, `day_of_week`, `is_weekend`, and `origin_loc_id`. Since actual future demand values are unavailable during inference, the model does not rely on future lagged observations. This avoids recursive error accumulation but may result in smoother predictions when unexpected demand shocks occur.

4. Deployment Limitation

Batch-Based Forecast Generation:

The current deployment generates the demand forecast as a batch output (`demand_forecast.csv`) before visualization in Tableau. Therefore, the system is designed primarily for periodic operational forecasting rather than continuous real-time streaming from live taxi telemetry.

15.Conclusion

15. CONCLUSION

This study presented an end-to-end **Urban Flow Analytics** solution for transforming large-scale taxi trip data into reliable analytical insights, predictive outputs, and operational decision support. The developed pipeline addressed key data-quality challenges through systematic validation, chunk-based processing, feature engineering, encoding, scaling, and leakage-aware temporal processing. These preprocessing stages established a consistent analytical foundation for subsequent predictive modelling and spatial-temporal analysis.

The predictive components demonstrated the value of machine learning for urban transportation operations. The **XGBoost-based Trip Duration Estimator** achieved an R^2 of **0.8248**, with an RMSE of 647.38 seconds and MAE of 203.82 seconds on the validation set. For demand forecasting, the XGBoost model achieved an **RMSE of 14.90, MAE of 6.46, and R^2 of 0.9575**, explaining approximately 95.75% of the variation in hourly zone-level pickup demand. The forecasting engine further generated demand predictions across **261 taxi zones for a 72-hour horizon**, providing a practical basis for proactive fleet positioning and operational planning.

In addition, the **Hotspot and Origin-Destination analysis** extended the solution from individual trip-level observations to broader spatial-temporal mobility patterns. The analysis incorporated hourly pickup and drop-off activity and OD flow information to identify concentrated demand and major movement patterns, while the optimized processing architecture addressed the computational challenges associated with large spatial flow structures.

Finally, the integration of the forecasting outputs with the **Tableau business dashboard** converted machine-learning predictions into an interpretable decision-support interface. Actual and predicted demand, peak-demand periods, and high-demand zones can be examined interactively, allowing fleet managers to move from historical analysis toward forward-looking operational planning. Therefore, the overall solution demonstrates how large-scale transportation data can be systematically transformed into **insights, predictions, and actionable business decisions**, supporting more proactive fleet allocation, demand planning, operational efficiency, and improved customer service. While the current system remains limited by factors such as the absence of external real-time variables and batch-based forecast generation, it provides a strong foundation for future enhancements involving richer contextual data and more real-time operational integration.

16. References

- [1] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, San Francisco, CA, USA, 2016, pp. 785–794.
- [2] L. Moreira-Matias, J. Gama, M. Ferreira, J. Mendes-Moreira, and L. Damas, “Predicting Taxi–Passenger Demand Using Streaming Data to Enable Dynamic Resource Allocation,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 14, no. 3, pp. 1393–1404, 2013.
- [3] J. Zhang, Y. Zheng, and D. Qi, “Deep Spatio-Temporal Residual Networks for Citywide Crowd Flows Prediction,” in *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence (AAAI-17)*, San Francisco, CA, USA, 2017, pp. 1655–1661.
- [4] D. Wang, J. Zhang, W. Cao, J. Li, and Y. Zheng, “When Will You Arrive? Estimating Travel Time Based on Deep Neural Networks,” in *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI-18)*, New Orleans, LA, USA, 2018.
- [5] E. Cascetta, “Estimation of Trip Matrices from Traffic Counts and Survey Data: A Generalized Least Squares Approach,” *Transportation Research Part B: Methodological*, vol. 18, nos. 4–5, pp. 289–299, 1984.
- [6] New York City Taxi and Limousine Commission, “TLC Trip Record Data,” New York City, NY, USA. [Online]. Available: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- [7] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [8] XGBoost Developers, “XGBoost Documentation,” XGBoost Documentation. [Online]. Available: <https://xgboost.readthedocs.io/>
- [9] Tableau Software, “Tableau Documentation,” Salesforce, Tableau. [Online]. Available: <https://help.tableau.com/>
- [10] SLIIT Codefest 2026, “Urban Flow Analytics Data Challenge,” Data Challenge Brief and Dataset Documentation, 2026.